

Tutorial Sheet-03

1. Let X be a continuous random variable with the probability density function (PDF):

$$f(x) = \begin{cases} \frac{2}{x^3}, & x \geq 1 \\ 0, & x < 1 \end{cases}$$

- (a) Show that the expectation $E(X)$ exists and find its value.
- (b) Show that the second moment $E(X^2)$ does not exist. What does this imply about the variance and higher-order moments of X ?

2. A discrete random variable X has the probability mass function:

$$P(X = k) = \frac{6}{\pi^2 k^2}, \quad k = 1, 2, 3, \dots$$

Does the expected value $E(X)$ exist? Justify your answer. Give an example where $\sum xp_X(x)$ converges but $\sum |x|p_X(x)$ diverges. Does $E(X)$ exist? Explain briefly.

3. Let X be a random variable with PDF $f(x) = 2x$ for $0 < x < 1$. Find the expected value of the random variable $Y = e^X$.
4. Let X be a continuous random variable with PDF:

$$f(x) = Ce^{-(x-2)^2}, \quad -\infty < x < \infty$$

- (a) Identify the center of symmetry a for this distribution.
 - (b) Without integration, state the values of $E(X)$ and $E(X - 2)^3$.
5. Using the linearity of expectation, derive the relationship between the third central moment $\mu_3 = E[(X - \mu)^3]$ and the raw moments μ'_1, μ'_2, μ'_3 .
6. Let X be a random variable with $E(X) = 2$ and $E(X^2) = 8$.
- (a) Calculate the variance σ^2 and Standard Deviation σ .
 - (b) Calculate $E(3X - 5)$ and $Var(3X - 5)$.
7. Show $Var(X) = 0$ iff X is degenerate.
8. Let X be a random variable with the exponential CDF:

$$F(x) = \begin{cases} 1 - e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- (a) Find the median m of the distribution.
 - (b) Find the quantile of order $p = 0.75$ (the 75th percentile).
9. Let X be a random variable with parameter p such that $P(X = 1) = p, P(X = 0) = 1 - p$.
- (a) Find the Probability Generating Function (PGF) $P(s)$.
 - (b) Use the PGF to find the mean $E(X)$ using the derivative $P'(1)$.

10. The PGF of a random variable X is given by $P(s) = e^{\lambda(s-1)}$.
- (a) Find the first and second derivatives $P'(s)$ and $P''(s)$.
 - (b) Use these to calculate $E(X)$ and $Var(X)$.
11. Find the Moment Generating Function $M_X(t)$ for the random variable X with PDF:

$$f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

For what range of t does $M_X(t)$ exist?

Let X be a random variable with MGF $M_X(t)$. Let $Y = aX + b$. Show that the MGF of Y is given by:

$$M_Y(t) = e^{bt} M_X(at)$$

12. The MGF of a random variable X has the Maclaurin series expansion:

$$M_X(t) = 1 + 3t + \frac{14t^2}{2!} + \dots$$

Identify the values of:

- (a) The first moment $\mu'_1 = E(X)$.
 - (b) The second moment $\mu'_2 = E(X^2)$.
 - (c) The variance $Var(X)$.
13. Consider the random variable X with PDF $f(x) = \frac{1}{2}e^{-|x|}$ for $-\infty < x < \infty$ (Laplace Distribution).
- (a) Find the MGF $M_X(t)$ and state the interval of convergence.
 - (b) Explain why the MGF does not exist for all $t \in \mathbb{R}$, whereas moments of all orders exist for this distribution.